HARD · ADVANCED MATH

Nonlinear functions

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

A quadratic function models a projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. The model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. How many seconds after the launch does the model estimate that the projectile will return to a height of 7 meters?

A. 3

B. 6

C. 7

D. 9

Q2

ADVANCED MATH

$$P t = 2601.04^{\frac{6}{4} t}$$

The function P models the population, in thousands, of a certain city t years after 2003. According to the model, the population is predicted to increase by 4% every n months. What is the value of n?

A. 8

B. 12

C. 18

D. 72

Q3

GRID-IN ADVANCED MATH

The quadratic function g models the depth, in meters, below the surface of the water of a seal t minutes after the seal entered the water during a dive. The function estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water and then reached the surface of the water 12 minutes after it entered the water. Based on the function, what was the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ADVANCED MATH

The first term of a sequence is 9. Each term after the first is 4 times the preceding term. If w represents the nth term of the sequence, which equation gives w in terms of n?

A. $w = 4 9^n$

B. $w = 4 9^{n-1}$

C. $w = 9 4^n$

D. $w = 9 4^{n-1}$

Q5

ADVANCED MATH

$$Pt = 2901.04^{\frac{4}{6}t}$$

The function P models the population, in thousands, of a certain city t years after 2005. According to the model, the population is predicted to increase by $n\%$ every 18 months. What is the value of n?

A. 0.38

B. 1.04

C. 4

D. 6

Q6

ADVANCED MATH

$$f x = 59 - 2x$$

The function f is defined by the given equation. For which of the following values of k does $f k = 3k$?

A. $\frac{59}{5}$

B. $\frac{59}{2}$

C. $\frac{177}{5}$

D. 59

Q7 GRID-IN ADVANCED MATH

The product of two positive integers is 462. If the first integer is 5 greater than twice the second integer, what is the smaller of the two integers?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN ADVANCED MATH

$$f x = x^2 - 48x + 2{,}304$$

What is the minimum value of the given function?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 ADVANCED MATH

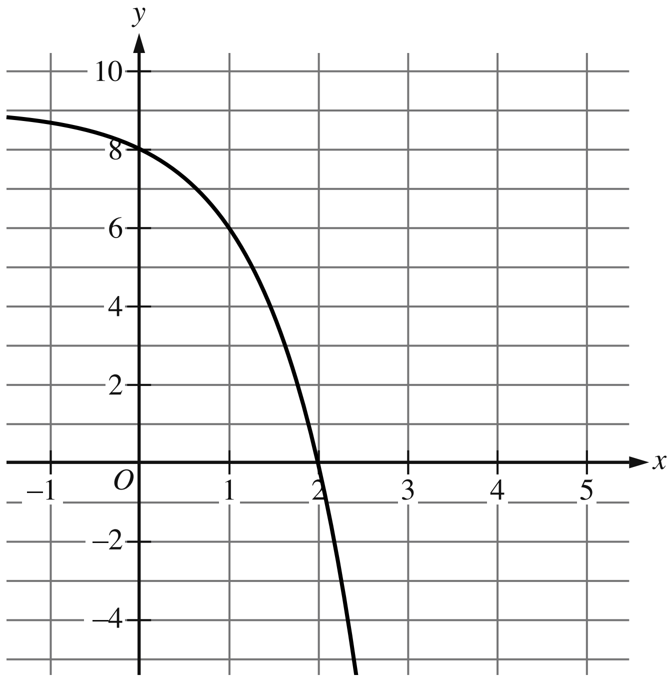
A park ranger hung squirrel houses each in the shape of a right rectangular prism for fox squirrels. Each house has a height of 11 inches. The length of each house's base is x inches, which is 1 inch more than the width of the house's base. Which function V gives the volume of each house, in cubic inches, in terms of the length of the house's base?

A. $Vx = 11xx - 1$

B. $Vx = 11xx + 1$

C. $Vx = xx + 11x - 1$

D. $Vx = xx + 11x + 1$



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
  - In quadrant 2, the curve trends down gradually to the point (0 comma 8).
  - In quadrant 1:
    - The curve trends down gradually to the approximate point (1 comma 6.0).
    - The curve then trends down sharply to the x axis.
  - In quadrant 4, the curve trends down sharply.
- As x decreases, the curve approaches the line y equals 9.
- The curve passes through the following points:
  - (0 comma 8)
  - (1 comma 6)

The graph of $y = fx + 4$ is shown. Which equation defines function f?

A. $fx = \text{-}3^x + 1$

B. $f x = \text{-}3^x + 5$

C. $f x = \text{-}3^x + 8$

D. $f x = \text{-}3^x + 9$